

Kelvin Olonade

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EDUCATION

University of Lagos

B.Sc. Electrical & Electronics Engineering — CGPA: 4.18/5.00

Lagos, Nigeria

Expected Oct 2028

TECHNICAL SKILLS

HDL & Design: Verilog, SystemVerilog, RTL Design, FSM Design, Synthesizable IP, Parameterized Modules
Verification: Self-checking Testbenches, Functional Coverage, Constrained-Random, Assertions, Edge-Case Analysis, Cocotb
EDA & Flow: OpenLane, Yosys, Magic, KLayout, Icarus Verilog, GTKWave, Verilator, OpenROAD
Process & PDKs: SkyWater SKY130, GlobalFoundries GF180, Standard-Cell Synthesis, STA Basics, Place & Route, DRC, LVS
Embedded & Tools: PIC16F CLB, CIP, ARM Cortex-M, C, C++, Python, Git, Linux, RHEL, MATLAB, Simulink

SILICON & HARDWARE PROJECTS

RV32E FIR SoC with Hardware Accelerator — GF180, TTGF180B Shuttle | In Progress

- Designing a minimal RV32E RISC-V SoC featuring a custom 16-tap FIR hardware accelerator on GF180MCU, originally targeting the TTGF180B shuttle tapeout; currently adapting the memory architecture to interface with external SRAM after initial floorplanning revealed the synthesized RAM macros exceeded available tile area.
- Implemented the full open-source ASIC flow including RTL synthesis through Yosys, macro hardening via OpenLane, and gate-level simulation with SDF-annotated timing; verified FIR mathematical accuracy against a Python golden model with ± 1 LSB tolerance.
- Building bare-metal firmware in C and RISC-V assembly with custom startup code and linker script; validated complete CPU + FIR + UART chain in RTL simulation proving $\geq 4\times$ hardware acceleration over software FIR on the same core.

Autonomous VGA Algorithmic Synthesizer ASIC — SKY130, TTSKY26A Shuttle

- Designed a tape-out-ready digital ASIC for TTSKY26A on SkyWater 130 nm, generating autonomous VGA 640×480 visuals from on-chip fractal and LFSR engines with zero firmware and pure digital logic.
- Hand-optimized RTL for extreme area constraints (1 tile $\approx 160 \times 100 \mu\text{m}^2$); pipelined the pixel pipeline at 25.175 MHz and balanced critical paths through OpenLane synthesis and STA.
- Implemented a self-checking SystemVerilog testbench with functional coverage on sync timing, fractal correctness, and entropy quality before signoff.

Hardware EEPROM Wear-Leveling Controller — SKY130, TTSKY26B Shuttle

- Designed and taped out a compact, attack-resistant wear-leveling controller implementing the Start-Gap algorithm (used in 3D XPoint) to extend EEPROM/flash lifetime; verified with Cocotb and synthesized on SkyWater 130 nm via Tiny Tapeout SKY26B.
- Incorporated a 2-round Feistel address scrambler with LFSR key to defeat targeted wear attacks, featuring 8-bit per-block wear counters, bad-block retirement, and in-band telemetry for max-min wear skew and total write count.
- Achieved successful GDS generation with full place-and-route and functional simulation sign-off, proving a silicon-ready IP core with hardware-level security and reliability.

EXPERIENCE

Embedded Systems Engineer

Balancee Tech Solutions

May 2026 – Present

Ajah, Lagos

- Led end-to-end design of a custom fuel-pump embedded controller, encompassing schematic capture, MCU firmware, sensor signal-chain design, and field-test verification.
- Encountered persistent EEPROM endurance failures during field deployment due to frequent calibration-write cycles, sparking a personal research into memory wear mechanisms that directly brought about the design of a hardware wear-leveling controller later taped out on SkyWater 130 nm.

Embedded Systems Intern

AVZDAX

Dec 2025 – Present

Victoria Island, Lagos

- Delivered an edge CV pipeline on Pi 5 and Pi 4B, integrating a Hailo-8 AI accelerator for all inference workloads on a custom-trained proprietary model via hardware-software co-design while eliminating pipeline stalls through systematic bottleneck profiling across camera capture, pre-processing, and accelerator hand-off stages, achieving stable throughput well above the 30 FPS real-time target on the integrated Hailo-8 plus Pi 5 platform.

LEADERSHIP, CERTIFICATIONS & HOBBIES

Established the first IEEE Solid-State Circuits Society (SSCS) Student Chapter at the University of Lagos as Founding Chair, organizing RTL/ASIC workshops and building Nigeria's pipeline of silicon engineers.

Arm Developer Program Contributor — Leveraged technical documentation from the Arm developer community to design, troubleshoot and implement Cortex-M architecture solutions.

Certifications: Zero to ASIC Course, ARM Cortex-M Processors Overview amongst others.

Hobbies: Playing racing games, writing and making plans with intense research for the fun of it.